The documentation and process conversion measures necessary to comply with this document shall be completed by 11 October 2023.

INCH-POUND

MIL-PRF-19500/399G
w/AMENDMENT 2
11 July 2023
SUPERSEDING
MIL-PRF-19500/399G
w/AMENDMENT 1
14 March 2018

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, NPN, SILICON, SWITCHING
TYPE 2N3960 AND 2N3960UB
JAN, JANTX, JANTXV, JANS, JANHC, JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN silicon, switching transistors. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each device type and two levels for unencapsulated die as specified in MIL-PRF-19500. Two levels of product assurance (JANHC and JANKC) are provided for each unencapsulated device type. RHA level designators “M”, “D”, “P“, “L”, “R”, “F”, “G”, and “H” are appended to the device prefix to identify devices which have passed RHA requirements.

1.2 Physical dimensions. The device packages for the encapsulated device types are as follows: (2N3960) (TO-18) in accordance with figure 1 and (UB) in accordance with figure 2. The dimensions and topography for JANHC and JANKC unencapsulated die is as follows: The A version die in accordance with figure 3.

1.3 Maximum ratings. Unless otherwise specified, T_C = +25°C.

P_T (1) T_A = +25°C	V_{CBO}	V_{CEO}	V_{EBO}	T_J and T_{STG}
mW	V dc	V dc	V dc	°C
400	20	12	4.5	-65 to +200

(1) Derate linearly 2.3 mW/°C above T_A = +25°C.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.4 Primary electrical characteristics.

Limits	h_{FE1} V_{CE} = 1.0 V dc I_C = 1.0 mA dc	h_{FE2} V_{CE} = 1.0 V dc I_C = 10 mA dc	$V_{CE(sat)1}$ I_C = 1.0 mA dc I_B = 0.1 mA dc	$V_{CE(sat)2}$ I_C = 30 mA dc I_B = 3.0 mA dc	C_{obo} V_{CB} = 4 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz	$\lvert h_{fe} \rvert$ V_{CE} = 4 V dc I_C = 5.0 mA dc f = 100 MHz	V_{BE1} I_C = 1.0 mA dc V_{CE} = 1.0 V dc	V_{BE2} I_C = 30 mA dc V_{CE} = 1.0 V dc
			V dc	V dc	pF		V dc	V dc
Min	40	60				13		
Max		300	0.2	0.3	2.5		0.8	1.0

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.5 for PIN construction example and 6.6 for a list of available PINs.

1.5.1 JAN certification mark and quality level.

1.5.1.1 Quality level designators for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", "JANTXV", and "JANS".

1.5.1.2 Quality level designators for unencapsulated devices (die). The quality level designators for unencapsulated devices (die) that are applicable for this specification sheet from the lowest to the highest level are as follows: "JANHC" and "JANKC".

1.5.2 Radiation hardness assurance (RHA) designator. The RHA levels that are applicable for this specification sheet from lowest to highest are as follows: "M", "D", "P", "L", "R", "F", "G", and "H").

1.5.3 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

1.5.3.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

1.5.3.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are as follows: "3960".

1.5.4 Suffix symbols. The following suffix letters are incorporated in the PIN in the order listed in the table as applicable:

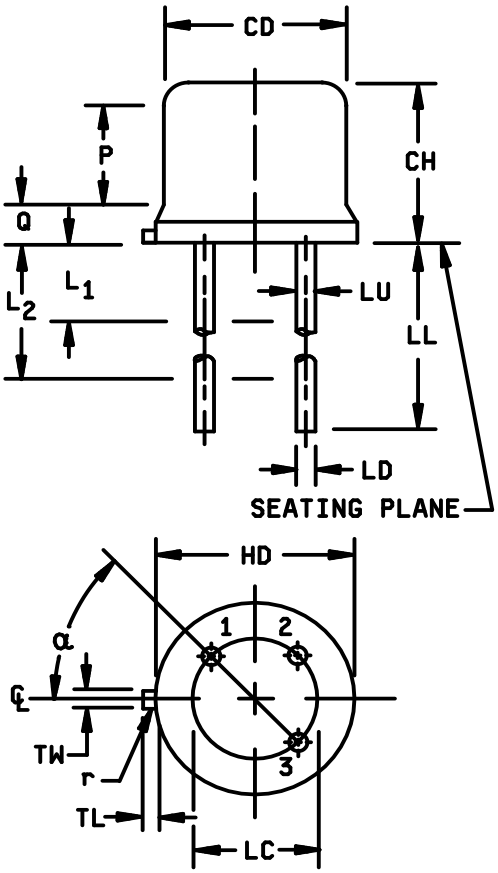
UB	Indicates a 4 pad surface mount package. The metal lid is connected to pad 4 (see figure 2)

1.5.5 Lead finish. The lead finishes applicable to this specification sheet are listed on QML-19500.

1.5.6 Die identifiers for unencapsulated devices (manufacturers and critical interface identifiers). The manufacturer die identifiers that are applicable for this specification sheet are "A"

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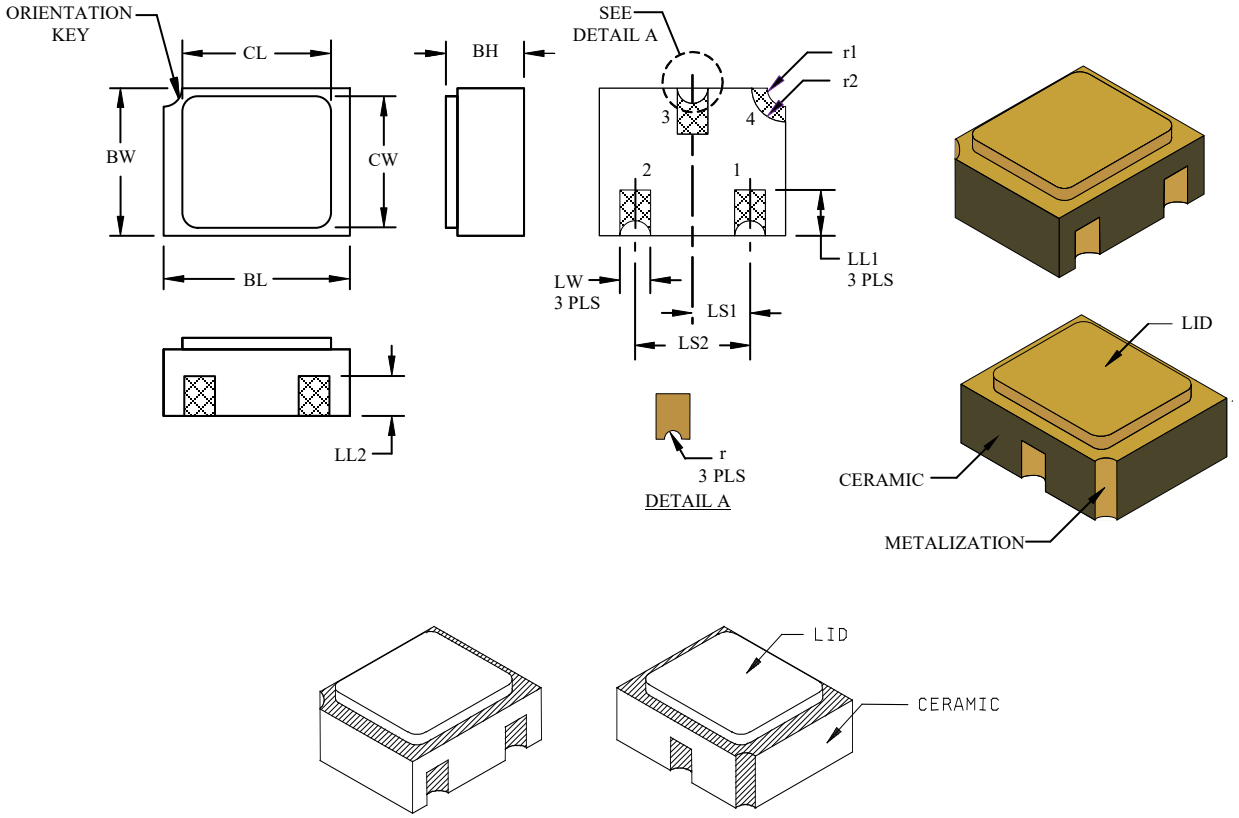
Symbol	Dimensions				Note
	in	in	mm	mm	
	Min	Max	Min	Max	
CD	.178	.195	4.52	4.95	
CH	.170	.210	4.32	5.33	
HD	.209	.230	5.31	5.84	
LC	.100 TP		2.54 TP		6
LD	.016	.021	0.41	0.53	7,11
LL	.500	.750	12.70	19.05	7
LU	.016	.019	0.41	0.48	12
L1		.050		1.27	7
L2	.250		6.35		7
P	.100		2.54		5
Q		.040		1.02	4
TL	.028	.048	0.71	1.22	3
TW	.036	.046	0.91	1.17	9
r		.010		0.25	10
α	45° TP		45° TP		6



NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Symbol TL is measured from HD maximum.
4. Details of outline in this zone are optional.
5. Symbol CD shall not vary more than .010 (0.25 mm) in zone P. This zone is controlled for automatic handling.
6. Leads at gauge plane .054 inch (1.37 mm) +.001 inch (0.03 mm) -.000 inch (0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) relative to tab. Device may be measured by direct methods or by gauge.
7. Symbol LD applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum.
8. Lead number three is electrically connected to case.
9. Beyond r maximum, TW shall be held for a minimum length of .011 inch (0.28 mm).
10. Symbol r applied to both inside corners of tab.
11. Measured in a zone beyond .250 (6.35 mm) from the seating plane.
12. Measured in the zone between .050 (1.27 mm) and .250 (6.35 mm) from the seating plane.
13. Lead 1 is emitter, lead 2 is base, and case is collector.

FIGURE 1. Physical dimensions (similar to TO-18).

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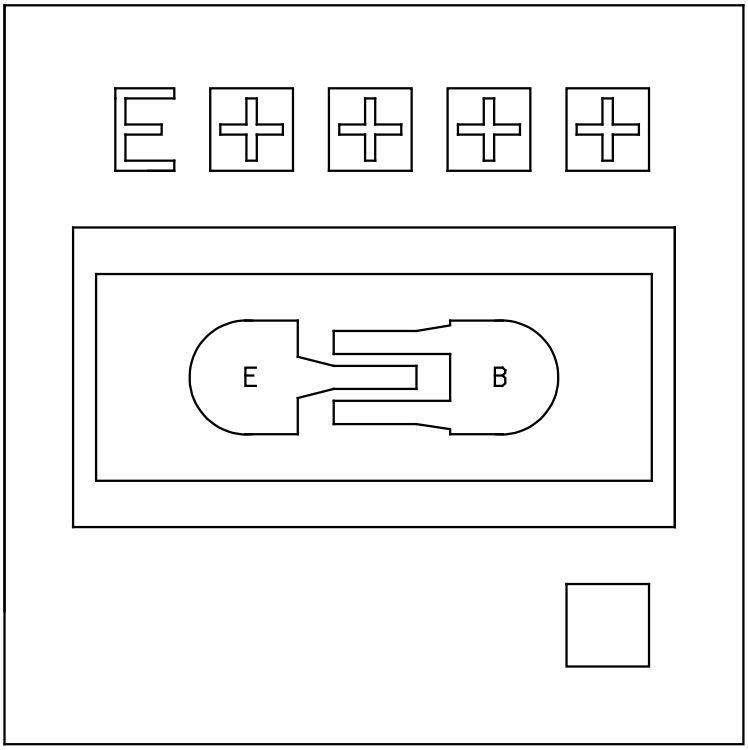


Ltr.	Dimensions				Note	Ltr.	Dimensions				Note
	in	in	mm	mm			in	in	mm	mm	
	Min	Max	Min	Max			Min	Max	Min	Max	
BH	.046	.056	1.17	1.42		LS1	.035	.040	0.89	1.02	
BL	.115	.128	2.92	3.25		LS2	.071	.079	1.80	2.01	
BW	.085	.108	2.16	2.74		LW	.016	.024	0.41	0.61	
CL		.128		3.25		r		.008		0.20	
CW		.108		2.74		r1		.012		0.31	
LL1	.022	.038	0.56	0.96		r2		.022		0.56	
LL2	.017	.035	0.43	0.89							

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Hatched areas on package denote metallized areas.
4. Lid material: Kovar.
5. Pad 1 = Base, Pad 2 = Emitter, Pad 3 = Collector, Pad 4 = Shielding connected to the lid.

FIGURE 2. Physical dimensions for 2N3960UB, surface mount.

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Die size:	.016 x .016 inch (0.41 x 0.41 mm).
Die thickness:	.008 ±.0016 inch (0.20 ±0.041 mm).
Base pad:	.0027 x .0027 inch 0.069 x 0.069 mm).
Emitter pad:	.0027 x .0027 inch.
Back metal:	Gold, 6500 ±1950 Ang.
Top metal:	Aluminum, 17500 ±2500 Ang.
Back side:	Collector.
Glassivation:	SiO_2, 7500 ±1500 Ang.

FIGURE 3. JANHC and JANKC (A-version) die dimensions.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML-19500) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and figures 1 (TO-18), figure 2 (UB), and figure 3 (JANHC, JANKC).

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Radiation hardness assurance (RHA). Radiation hardness assurance requirements, PIN designators, and test levels shall be as defined in MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

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3.8 Marking. Devices shall be marked in accordance with MIL-PRF-19500, except for the UB suffix package. Marking on the UB package shall consist of an abbreviated part number, the date code, and the manufacturer's symbol or logo. The prefixes JAN, JANTX, JANTXV, and JANS can be abbreviated as J, JX, JV, and JS respectively. The "2N" prefix and the "UB" suffix can also be omitted. The radiation hardened designator M, D, P, L, R, F, G, or H shall immediately precede (or replace) the device “2N” identifier (depending upon degree of abbreviation required).

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and table I, table II, table III).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of Table IV tests, the tests specified in table IV herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.2.2 JANHC and JANKC qualification. JANHC and JANKC qualification inspection shall be in accordance with MIL-PRF-19500.

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4.3 Screening (JANS, JANTX and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurements for JANS level	Measurements for JANTX and JANTXV levels
3c	Thermal impedance Method 3131 of MIL-STD-750 (see 4.3.3).	Thermal impedance Method 3131 of MIL-STD-750 (see 4.3.3). (1)
9	I_{CEX2} and h_{FE2}	Not applicable
10	V_{CB} = 12 V, 48 hours minimum.	V_{CB} = 12 V, 48 hours minimum.
11	I_{CEX2} and h_{FE2} ΔI_{CEX2} = 100 percent of initial value or 2 nA dc, whichever is greater; Δh_{FE2} = ±20 percent.	I_{CEX2} and h_{FE2}
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; ΔI_{CEX2} = 100 percent of initial value or 2 nA dc, whichever is greater; Δh_{FE2} = ±20 percent.	Subgroup 2 of table I herein; ΔI_{CEX2} = 100 percent of initial value or 2 nA dc, whichever is greater; Δh_{FE2} = ±20 percent.
14	Required.	Required.

(1) Shall be performed anytime after temperature cycling, screen 3a; and does not need to be repeated in screening requirements for JANTX and JANTXV level devices.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: V_{CB} = 10 V dc. Power shall be applied to achieve T_J = +135°C minimum using a minimum P_D = 75 percent of P_T maximum. T_A = room ambient as defined in 1.3. With approval of the qualifying activity and preparing activity, alternate burn-in data criteria (hours, bias conditions, T_J, and mounting conditions) may be used for JANTX and JANTXV quality levels. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval. Method 3100 of MIL-STD-750 to measure T_J shall be used.

4.3.2 Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with MIL-PRF-19500, "Discrete Semiconductor Die/Chip Lot Acceptance". Burn-in duration for the JANKC level follows JANS requirements; the JANHC follows JANTX requirements.

4.3.3 Thermal impedance. The thermal impedance shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining V_H, V_{CE}, I_M, I_H, t_H, and t_{MD}. The limit used in screen 3c and table I, subgroup 2 shall be set statistically by the supplier over several die lots and submitted to the qualifying activity for approval.

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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein. If alternate screening is being performed in accordance with MIL-PRF-19500, a sample of screened * devices shall be submitted to, and pass, the requirements of group A1 and A2 inspection only (table E-VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-VIa (JANS) of MIL-PRF-19500 and 4.4.2.1 herein. (See 4.4.2.2 for JAN, JANTX, and JANTXV group B testing). Delta requirements shall be in accordance with table III herein as specified in the footnotes for table III.

4.4.2.1 Group B inspection, table E-VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Conditions
B4	1037	V_{CB} = 10 V dc, 2,000 cycles, t_{ON} = t_{OFF} = 3 minutes, $P_{D(ON)}$ = P_D max rated per 1.3; $P_{D(OFF)}$ = 0.
B5	1027	V_{CB} = 10 V dc, 1,000 hours at 75 percent of maximum rated power shall be applied and ambient temperature adjusted to achieve T_J = +150°C minimum. n = 45, c = 0.

4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV). Separate samples may be used for each step. In the event of a group B failure, the manufacturer may pull a new sample at double size from either the failed assembly lot or from another assembly lot from the same wafer lot. If the new “assembly lot” option is exercised, the failed assembly lot shall be scrapped.

Step	Method	Condition
1	1026	Steady-state life: Test condition B, 1000 hours minimum, V_{CB} = 10 V dc, power shall be applied to the device and T_A adjusted to achieve T_J = +150°C minimum, and minimum P_D = 75 percent of max rated P_T (see 1.3 herein); n = 45, c = 0.
2	1048	HTRB: Test condition A, 48 hours minimum. n = 45, c = 0.
3	1032	High- temperature life (non-operating), T_A = +200°C, t = 340 hours, n = 22, c = 0.

4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. For JANS samples shall be selected from each inspection lot. See MIL-PRF-19500.

b. Shall be chosen from an inspection lot that has been submitted to and passed group A, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B4 and B5 for JANS, and group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500 and as follows. Delta requirements in accordance with the notes of table III herein.

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4.4.3.1 Group C inspection, table E-VII (JANS) of MIL-PRF-19500.

Subgroup	Method	Conditions
C2	2036	Test condition E, not applicable to UB.
C6	1026	V_{CB} = 10 V dc, 1,000 hours; maximum rated power shall be applied and T_A adjusted to achieve T_J = +150°C minimum and a minimum of P_D = 75 percent of maximum rated PT as defined in 1.3.

4.4.3.2 Group C inspection, table VII (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E, (method 2036 not applicable for UB devices).
C6	1026	Not applicable.

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes group A tests for conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for C6 life test may be pulled prior to the application of final lead finish. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.4 Group D inspection. Conformance inspection for hardness assured JANS and JANTXV types shall include the group D tests specified in table II herein. These tests shall be performed as required in accordance with MIL-PRF-19500 and method 1019 of MIL-STD-750 for total ionizing dose, or method 1017 of MIL-STD-750 for neutron fluence, as applicable (see 6.2 herein), except group D, subgroup 2 may be performed separate from other subgroups. Alternate package options may also be substituted for the testing provided there is no adverse effect to the fluence profile.

4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table IV herein. Delta requirements shall be in * accordance with the notes of table III herein.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Displacement damage characterization. For RHA devices, each supplier shall perform a displacement damage characterization. The characterization shall demonstrate exposure versus data and does not indicate pass or fail criteria. The exposure shall be conducted in accordance with method 1017 of MIL-STD-750. The following details shall apply:

a. Samples may be taken from any wafer of the qualification lot.

b. As a minimum, testing shall be at 2E+12 n/cm^2 plus two additional fluence levels chosen by the manufacturer.

c. If the device degrades less than 5 percent of the specification at the highest fluence level, a single data point may be sufficient.

Alternate package options may be substituted for characterization. The displacement damage characterization data shall be made available from the manufacturer.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits	Limits	Unit
	Method	Conditions		Min	Max	
Subgroup 1 2/						
Visual and mechanical inspection 3/	2071					
Solderability 3/ 4/	2026	n = 15 leads, c = 0				
Resistance to solvents 3/ 4/ 5/	1022	n = 15 devices, c = 0				
Salt atmosphere (corrosion) 4/	1041	n = 6 devices, c = 0, (For laser marked devices only)				
Temp cycling 3/ 4/	1051	Test condition C, 25 cycles. n = 22 devices, c = 0				
Hermetic seal 4/ Fine leak Gross leak	1071	n = 22 devices, c = 0				
Electrical measurements 4/		Table I, subgroup 2				
Bond strength 3/ 4/	2037	Precondition T_A = +250°C at t = 24 hrs or T_A = +300°C at t = 2 hrs n = 11 wires, c = 0				
De-cap internal visual	2075	n = 4, c = 0				
Subgroup 2						
Thermal impedance 6/	3131	See 4.3.3	$Z_{\theta JX}$			mV
Collector to base cutoff current	3036	Bias condition D, V_{CB} = 20 V dc	I_{CBO}		10	µA dc
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 4.5 V dc	I_{EBO}		10	µA dc
Breakdown voltage, collector to emitter	3011	Bias condition D, I_C = 10 µA dc, pulsed (see 4.5.1)	$V_{(BR)CEO}$	12		V dc
* Collector to emitter cutoff current	3041	Bias condition A; V_{CE} = 10 V dc, V_{EB} = 0.4 V dc	I_{CEX1}		1.0	µA dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

	Inspection 1/	MIL-STD-750		Symbol	Limits	Limits	Unit
		Method	Conditions		Min	Max	
	Subgroup 2 - Continued						
*	Collector to emitter cutoff current	3041	Bias condition A; V_{CE} = 10 V dc, V_{EB} = 2.0 V dc	I_{CEX2}		5.0	nA dc
	Forward-current transfer ratio	3076	V_{CE} = 1 V dc; I_C = 1.0 mA dc	h_{FE1}	40		
	Forward-current transfer ratio	3076	V_{CE} = 1 V dc; I_C = 10 mA dc, pulsed (see 4.5.1)	h_{FE2}	60	300	
	Forward-current transfer ratio	3076	V_{CE} = 1 V dc; I_C = 30 mA dc; pulsed (see 4.5.1)	h_{FE3}	30		
	Collector to emitter saturated voltage	3071	I_C = 1.0 mA dc; I_B = 0.1 mA dc	$V_{CE(sat)1}$		0.2	V dc
	Collector to emitter saturated voltage	3071	I_C = 30 mA dc; I_B = 3.0 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)2}$		0.3	V dc
	Base emitter voltage (non-saturated)	3066	Test condition B, V_{CE} = 1.0 V dc, I_C = 1.0 mA dc	V_{BE1}		0.8	V dc
	Base emitter voltage (non-saturated)	3066	Test condition B, V_{CE} = 1.0 V dc, I_C = 30 mA dc	V_{BE2}		1.0	V dc
	Subgroup 3						
	High-temperature operation:		T_A = +150°C				
	Collector to emitter cutoff current	3041	Bias condition A; V_{CE} = 10 V dc, V_{EB} = 2 V dc	I_{CEX3}		5.0	µA dc
	Low-temperature operation:		T_A = -55°C				
	Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 10 mA dc; pulsed (see 4.5.1)	h_{FE4}	30		

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits	Limits	Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Magnitude of common emitter small-signal short- circuit forward-current transfer ratio	3306	V_{CE} = 4 V dc; I_C = 5.0 mA dc; f = 100 MHz	$\|h_{fe1}\|$	13		
Magnitude of common emitter small-signal short- circuit forward-current transfer ratio	3306	V_{CE} = 4 V dc; I_C = 10 mA dc; f = 100 MHz	$\|h_{fe2}\|$	14		
Magnitude of common emitter small-signal short- circuit forward-current transfer ratio	3306	V_{CE} = 4 V dc; I_C = 30 mA dc; f = 100 MHz	$\|h_{fe3}\|$	12		
Open circuit output capacitance	3236	V_{CB} = 4 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		2.5	pF
Input capacitance (output open-circuited)	3240	V_{EB} = 0.5 V, I_C = 0, 100 kHz ≤ f ≤ 1 MHz	C_{ibo}		2.5	pF
Subgroups 5, 6, and 7						
Not applicable						

1/ For sampling plan, unless otherwise specified, see MIL-PRF-19500.
2/ For resubmission of failed subgroup A1, double the sample size of the failed test or sequence of tests. A failure in group A, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for JANS devices
5/ Not required for laser marked devices.
6/ For end-point measurements, this test is required for the following subgroups:
Group B, step 1 of 4.4.2.2 herein (JAN, JANTX, and JANTXV).
Group B, subgroups 3, 4, and 5 (JANS).
Group C, subgroup 2 and 6.
Group E, subgroup 1 and 2.

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* TABLE II. Group D inspection and end-point limits.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Limits	Limits	Unit
	Method	Conditions		Min	Max	
Subgroup 1 4/						
Neutron Irradiation	1017	Neutron exposure V_{CES} = 0 V				
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 10 uA dc, pulsed (see 4.5.1)	$V_{(BR)CEO}$	12		V dc
Collector to base cutoff current	3036	Bias condition D, V_{CB} = 20 V	I_{CBO}		20	uA dc
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 4.5 V dc	I_{EBO}		20	µA dc
* Collector to emitter cutoff current	3041	Bias condition A; V_{CE} = 10 V dc, V_{EB} = 0.4 V dc	I_{CEX1}		2.0	µA dc
* Collector to emitter cutoff current	3041	Bias condition A; V_{CE} = 10 V dc, V_{EB} = 2.0 V dc	I_{CEX2}		10	nA dc
Forward-current transfer ratio	3076	V_{CE} = 1 V dc, I_C = 1.0 mA dc	[h_{FE1}]	[20] 5/		
Forward-current transfer ratio	3076	V_{CE} = 1 V dc, I_C = 10 mA dc, pulsed (see 4.5.1)	[h_{FE2}]	[30] 5/	300	
Forward-current transfer ratio	3076	V_{CE} = 1 V dc, I_C = 30 mA dc, pulsed (see 4.5.1)	[h_{FE3}]	[15] 5/		
Collector to emitter saturated voltage	3071	I_C = 1.0 mA dc; I_B = 0.1 mA dc	$V_{CE(sat)1}$		0.23	V dc
Collector to emitter saturated voltage	3071	I_C = 30 mA dc; I_B = 3 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)2}$		0.35	V dc
Base emitter voltage (non saturated)	3066	Test condition B, V_{CE} = 1.0 V dc, I_C = 1.0 mA dc	V_{BE1}		0.92	V dc
Base emitter voltage (non saturated)	3066	Test condition B, V_{CE} = 1.0 V dc, I_C = 30 mA dc	V_{BE2}		1.15	V dc

See footnotes at end of table.

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TABLE II. Group D inspection and end-point limits – Continued.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Limits	Limits	Unit
	Method	Conditions		Min	Max	
Subgroup 2						
Steady-state total dose irradiation	1019	Gamma exposure V_{CES} = 9.6 V dc				
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 10 uA dc, pulsed (see 4.5.1)	$V_{(BR)CEO}$	12		V dc
Collector to base cutoff current	3036	Bias condition D, V_{CB} = 20 V	I_{CBO}		20	uA dc
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 4.5 V dc	I_{EBO}		20	µA dc
Collector to emitter cutoff current	3041	Bias condition A; V_{CE} = 10 V dc, V_{EB} = 0.4 V dc	I_{CEX1}		2.0	µA dc
* Collector to emitter cutoff current	3041	Bias condition A; V_{CE} = 10 V dc, V_{EB} = 2.0 V dc	I_{CEX2}		10	nA dc
* Forward-current transfer ratio	3076	V_{CE} = 1 V dc, I_C = 1.0 mA dc	[h_{FE1}]	[20] 5/		
Forward-current transfer ratio	3076	V_{CE} = 1 V dc, I_C = 10 mA dc, pulsed (see 4.5.1)	[h_{FE2}]	[30] 5/	300	
Forward-current transfer ratio	3076	V_{CE} = 1 V dc, I_C = 30 mA dc, pulsed (see 4.5.1)	[h_{FE3}]	[15] 5/		
Collector to emitter saturated voltage	3071	I_C = 1.0 mA dc; I_B = 0.1 mA dc	$V_{CE(sat)1}$		0.23	V dc
Collector to emitter saturated voltage	3071	I_C = 30 mA dc; I_B = 3 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)2}$		0.35	V dc
Base emitter voltage (non-saturated)	3066	Test condition B, V_{CE} = 1.0 V dc, I_C = 1.0 mA dc	V_{BE1}		0.92	V dc
Base emitter voltage (non-saturated)	3066	Test condition B, V_{CE} = 1.0 V dc, I_C = 30 mA dc	V_{BE2}		1.15	V dc

1/ Tests to be performed on all devices receiving radiation exposure.
2/ For sampling plan, see MIL-PRF-19500.
3/ Electrical characteristics apply to all device types unless otherwise noted.
4/ See 6.2.e herein.
5/ See method 1019 of MIL-STD-750, for how to determine [h_{FE}] by first calculating the delta(1/h_{FE}) from the pre- and post-radiation h_{FE}. Note that [h_{FE}] is not the same as h_{FE} and cannot be measured directly. The [h_{FE}] value can never exceed the pre-radiation minimum h_{FE} that it is based upon.

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TABLE III. Groups B, C, and E delta end-point inspection measurements. 1/ 2/ 3/ 4/

Step	Inspection	MIL-STD-750		Symbol	Limit	Limit	Unit
		Method	Conditions		Min	Max	
* 1.	Collector to emitter cutoff current	3041	Bias condition A, V_{CE} = 10 V dc, V_{EB} = 2.0 V dc	ΔI_{CEX2} 5/	100 percent of initial value or 2 nA dc, whichever is greater;		
2.	Forward current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 10 mA dc; pulsed (see 4.5.1)	Δh_{FE2} 5/	±20 percent change from initial value		

1/ The delta measurements for table VIa (JANS) of MIL-PRF-19500 are subgroup 4 and 5, see table III herein, all steps.
2/ The delta measurements for group B of 4.4.2.2 herein (JAN, JANTX, and JANTXV) are: Step 1, 2, and 3 of 4.4.2.2, see table III herein, all steps.
3/ The delta measurements for table E-VII of MIL-PRF-19500, Subgroup 6, all steps.
4/ The delta measurements for table E-IX of MIL-PRF-19500 are: Subgroups 1 and 2, see table III herein, all steps.
5/ Devices which exceed the table I limits herein for this test shall not be accepted.

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TABLE IV. Group E inspection (all quality levels) – for qualification or re-qualification only.

Inspection	MIL-STD-750		Qualification
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles.	
Hermetic seal Fine leak Gross leak	1071		
Electrical measurements		See table I, subgroup 2 and table III herein.	
Subgroup 2			45 devices c = 0
Intermittent life	1037	Intermittent operation life: V_{CB} = 10 V dc; 6,000 cycles; $t_{ON} = t_{OFF}$ = 3 minutes, $P_{D(ON)} = P_D$ max rated in accordance1.3; $P_{D(OFF)} = 0$.	
Electrical measurements		See table I, subgroup 2 and table III herein.	
Subgroup 4			Sample size N/A
Thermal impedance curves		See MIL-PRF-19500, group E, subgroup 4.	
Subgroup 5			
Not applicable			
Subgroup 6			11 devices
ESD	1020		
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition B	
* Subgroup 12			
Neutron Irradiation	1017	See 4.5.2	Neutron Irradiation

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete Part or Identifying Number (PIN), see 1.5.

e. For acquisition of RHA designed devices, table II, subgroup 1 testing of group D is optional. If subgroup 1 testing is desired, it should be specified in the contract.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML-19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

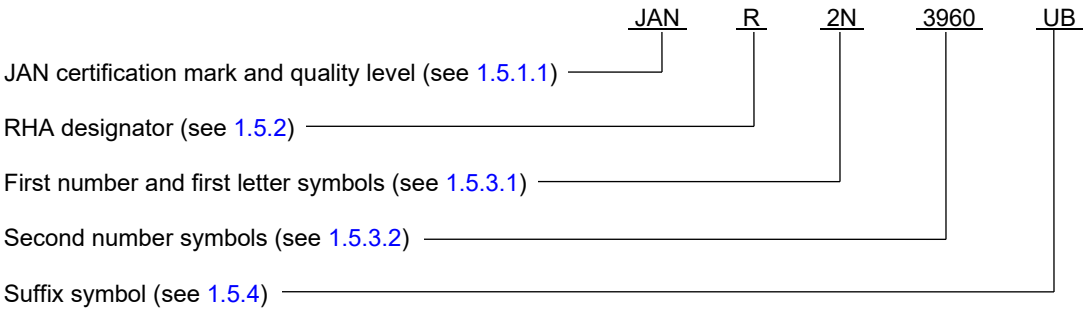
6.4 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCA2N3960) will be identified on the QML.

Die ordering information	
PIN	Manufacturer
	34156
2N3960	JANHCA2N3960 JANKCA2N3960

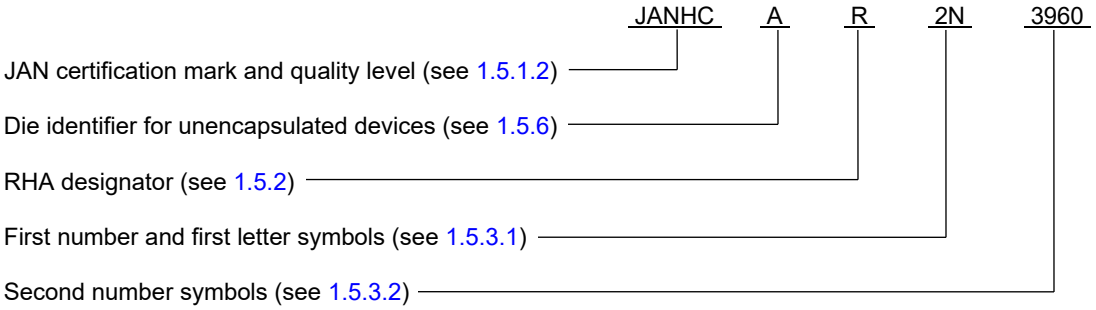
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6.5 PIN construction example.

6.5.1 Encapsulated devices The PINs for encapsulated devices are constructed using the following form.



6.5.2 Unencapsulated devices. The PINs for un-encapsulated devices are constructed using the following form.



6.6 List of PINs. The following is a list of possible PINs available on this specification sheet.

Encapsulated PINs for type 2N3960

JAN2N3960	JANTX2N3960	JANTXV2N3960	JANS#2N3960
JAN2N3960UB	JANTX2N3960UB	JANTXV2N3960UB	JANS#2N3960UB

Unencapsulated PINs for type 2N3960

JANHCA#2N3960	JANKCA#2N3960		

(1) The number sign (#) represent one of eight RHA designators available (M, D, P, L, R, F, G, or H). The PIN is also available without a RHA designator.

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6.7 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
* Navy - SH
Air Force - 85
NASA - NA
DLA - CC

Review activities:
Army - AR, MI
Navy - MC
Air Force - 19, 71

Preparing activity:
DLA - CC

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